

Assignment 3

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1 Uniformly Distributed Points in High-Dimensional spaces

1.1 Explanation of Code

generate_dataset(num_points, d):

Generates a random dataset with a specified number of points (num_points) and dimensionality (d), rounding each element to five decimal places.

compute_distances(query_point, dataset):

Computes L_1 , L_2 , and L_∞ distances between a query point and all points in a dataset, returning arrays for each distance metric.

compute_avg_ratios(nearest_distances, farthest_distances):

Computes the average ratio of farthest distances to nearest distances based on two arrays of distances.

main():

The main function that orchestrates the entire process. It generates datasets for various dimensions, randomly selects query points, computes distances, and calculates average ratios for L_1 , L_2 , and L_∞ norms. Finally, it plots the results with the y-axis in logarithmic scale and saves the plot as an image file.

1.2 Results

The plot visualizes the relationship between the dimensionality of a dataset and the average ratio of the farthest distances to nearest distances for three different distance metrics: L_1 , L_2 , and L_∞ norms.

Trend: As the dimensionality increases, you observe a decrease in the average ratio of farthest to nearest distances for all three norms. This phenomenon is known as the "curse of dimensionality," where the increase in dimensionality leads to sparsity in the data. As dimensions increase, points become more dispersed, concentrating mainly in the corners of the hypercube defined by the $[0, 1]$ range. This sparsity results in a rise in average distances between points.

Nearest distances increase due to wider point dispersion, while farthest distances reduce proportionally to the space's sparsity. This overall effect contributes to a decline in the average ratio of maximum to minimum distances.

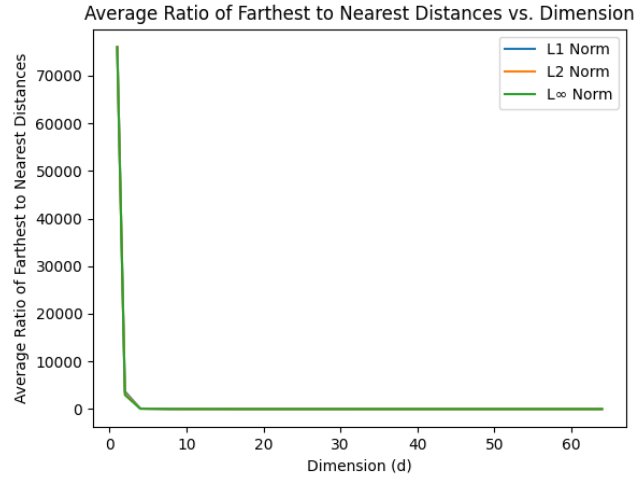


Figure 1: Volume vs dimension in normal scale

Since We cannot see clearly all the lines, a logarithmic scale is used for better presentation.

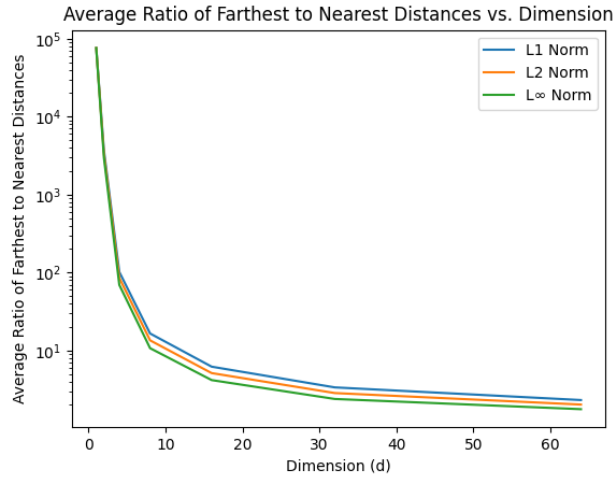


Figure 2: Volume vs dimension in logarithmic scale